

ACTL3142 Tutorial

Week 5: Generalised Linear Models

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Recap: Where We Left Off

From Weeks 2–3 (Linear Regression):

- ▶ Model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
- ▶ Key assumptions: $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, linearity, constant variance
- ▶ Model selection via AIC, BIC, adjusted R^2 , stepwise methods

The problem: linear regression assumes a Normal response with constant variance. In insurance and many other settings, this doesn't hold.

Today: **Generalised Linear Models**, a flexible framework that handles non-Normal responses.

Why Not Just Use OLS?

Think about typical insurance data:

- ▶ **Claim counts** are non-negative integers (0, 1, 2, ...), not Normal
- ▶ **Claim amounts** are positive and right-skewed, not Normal
- ▶ **Claim occurrence** is binary (0 or 1), definitely not Normal

OLS can give negative predictions for counts or amounts, and the constant-variance assumption is usually violated.

Multiplicative effects

In insurance pricing, effects are often **multiplicative** (e.g. young drivers cost $1.5\times$ more). GLMs with a log link model this naturally: $\ln \mu = \mathbf{x}'\boldsymbol{\beta}$, so $\mu = e^{\beta_0} \cdot e^{\beta_1 x_1} \dots$

The Three Components of a GLM (Q2)

Every GLM has three ingredients:

1. **Systematic component** (linear predictor):

$$\eta_i = \mathbf{x}_i' \boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$$

2. **Stochastic component** (distribution):

$$Y_i \mid \mathbf{X} \sim \text{EDF (Exponential Dispersion Family)}$$

3. **Link function** connecting them:

$$g(\mu_i) = \eta_i, \quad \text{i.e.} \quad g(\mathbb{E}[Y_i \mid \mathbf{X}]) = \mathbf{x}_i' \boldsymbol{\beta}$$

OLS is just a special case: Normal distribution + identity link
 $g(\mu) = \mu$.

The Exponential Dispersion Family (EDF)

The stochastic component requires Y to belong to the EDF. A distribution is in the EDF if its density can be written as:

$$f_Y(y; \theta, \psi) = \exp \left[\frac{y\theta - b(\theta)}{\psi} + c(y; \psi) \right]$$

- ▶ θ = **canonical parameter** (related to the mean)
- ▶ ψ = **dispersion parameter** (controls spread)
- ▶ $b(\theta)$ = cumulant function (determines the distribution)
- ▶ $c(y; \psi)$ = normalising term (doesn't depend on θ)

To show a distribution is in the EDF, take its density and rearrange into this form. That's exactly what Q3 and Q4(a) ask you to do.

Key EDF Properties

From the EDF form, two results drop out:

Mean:

$$E[Y] = b'(\theta) = \mu$$

Variance:

$$\text{Var}(Y) = \psi b''(\theta) = \psi V(\mu)$$

where $V(\mu) = b''(\theta)$ is the **variance function**.

Why this matters

The variance function $V(\mu)$ tells you how the variance relates to the mean. For Poisson, $V(\mu) = \mu$: higher mean $\Rightarrow$ higher variance. This is built into the model, unlike OLS which assumes constant variance.

Common EDF Members

Dist.	θ	$b(\theta)$	ψ	$V(\mu)$	Canon. link
Normal	μ	$\theta^2/2$	σ^2	1	Identity
Poisson	$\ln \mu$	e^θ	1	μ	Log
Gamma	$-1/\mu$	$-\ln(-\theta)$	$1/\nu$	μ^2	Reciprocal
Inv. Gauss.	$\frac{-1}{2\mu^2}$	$-\sqrt{-2\theta}$	$1/\lambda$	μ^3	$1/\mu^2$
Binomial	$\ln \frac{p}{1-p}$	$m \ln(1 + e^\theta)$	1	$\mu(1 - \mu/m)$	Logit

You don't need to memorise this, but you do need to be able to derive these from the density (Q3, Q4a, Q6). The variance function column is the most useful for understanding each distribution's behaviour.

Q3: Inverse Gaussian

★ [Solution](#) Consider the Inverse Gaussian distribution, as parametrised on Wikipedia (other parametrisations exist), i.e., with density:

3.
$$f(y) = \sqrt{\frac{\lambda}{2\pi y^3}} \exp\left(-\frac{\lambda(y - \mu)^2}{2\mu^2 y}\right), \quad y > 0.$$

- a. Show that this is a member of the exponential dispersion family. Hence, identify ψ and $b(\theta)$. **Hint:** you can rely on your knowledge that $\theta = -1/2\mu^2$ (as seen in the lecture).
- b. Find the mean and variance of the Inverse Gaussian (as function of μ and λ). You should rely on the properties of the exponential dispersion family, rather than doing tedious calculations ;-).

Q3: Showing Inverse Gaussian is in the EDF

Start from the IG density and expand the exponent:

$$f(y) = \sqrt{\frac{\lambda}{2\pi y^3}} \exp\left[-\frac{\lambda(y - \mu)^2}{2\mu^2 y}\right]$$
$$-\frac{\lambda(y^2 - 2y\mu + \mu^2)}{2\mu^2 y} = \underbrace{-\frac{\lambda y}{2\mu^2} + \frac{\lambda}{\mu}}_{\text{depends on } \mu} \underbrace{-\frac{\lambda}{2y}}_{\text{goes into } c}$$

Now rewrite the μ -dependent terms:

$$-\frac{\lambda y}{2\mu^2} + \frac{\lambda}{\mu} = \frac{1}{1/\lambda} \left[y \left(\frac{-1}{2\mu^2} \right) - \left(\frac{-1}{\mu} \right) \right] = \frac{y\theta - b(\theta)}{\psi}$$

So: $\theta = -1/(2\mu^2)$, $\psi = 1/\lambda$, $b(\theta) = -\sqrt{-2\theta}$.

Then $E[Y] = b'(\theta) = (-2\theta)^{-1/2} = \mu$ and

$\text{Var}(Y) = \psi b''(\theta) = \mu^3/\lambda$.

What Is the Link Function?

The link function g is just a transformation that connects the mean μ to the linear predictor:

$$g(\mu_i) = \eta_i = \mathbf{x}_i' \boldsymbol{\beta}$$

The identity link is $g(\mu) = \mu$, which means $\mu = \mathbf{x}'\boldsymbol{\beta}$ directly, no transformation at all. This is exactly what OLS does:

$$E[Y_i | \mathbf{X}] = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$$

So OLS is literally a GLM with Normal distribution + identity link.

Why does this work for Normal? Because μ can be any real number, and $\eta = \mathbf{x}'\boldsymbol{\beta}$ can be any real number, so there's no mismatch. But if μ must be **positive** (claims) or between **0 and 1** (probabilities), the identity link breaks down and we need a transformation.

Choosing a Link Function

The key question: **what range can μ take?** Since $\eta = \mathbf{x}'\beta \in (-\infty, +\infty)$, the link must map the range of μ onto this.

Range of μ	Link	$g(\mu)$	Effect type
$(-\infty, +\infty)$	Identity	μ	Additive
$(0, +\infty)$	Log	$\ln \mu$	Multiplicative
$(0, 1)$	Logit	$\ln \frac{\mu}{1-\mu}$	Odds ratios

Practical guidelines:

- ▶ Counts (Poisson): **log**, ensures $\hat{\mu} > 0$
- ▶ Amounts (Gamma): **log**, positive predictions + multiplicative rating factors
- ▶ Binary (Binomial): **logit**, keeps $\hat{p} \in (0, 1)$
- ▶ Normal: **identity**, no constraint needed

*The **canonical link** ($g(\mu) = \theta$) simplifies the math but isn't always the best practical choice. For Gamma, the canonical link is $1/\mu$, but log is preferred in insurance.*

Q1: Gamma GLM Fitted Values

1. ★ [Solution](#) An insurance company tested for claim sizes under two factors, i.e. **CAR**, the insurance group into which the car was placed, and **AGE**, the age of the policyholder (i.e. two-way contingency table). It was assumed that the claim size y_i follows a gamma distribution, under a log-link function. Analysis of a set of data for which $n = 8$ provided the following output, in the SAS software:

Observation	Claim size	CAR type	Age group	Pred	Xbeta	Resdev
1	27	1	1	25.53	3.24	0.30
2	16	1	2	24.78	3.21	-1.90
3	36	1	1		3.41	1.03
4	45	1	2	38.09	3.64	1.11
5	38	2	1	40.85	3.71	-0.46
6	27	2	2	36.97	3.61	-1.73
7	14	2	1		2.45	0.69
8	6	2	2	14.59	2.68	-2.55

Calculate the fitted claim sizes missing in the table.

Q1: Computing Fitted Values

Setup: 8 claim observations with factors CAR (3 levels) and AGE (2 levels). Gamma response with log link. SAS gives $\hat{\eta}_i$ (the “Xbeta” column) but some fitted values are missing.

Method:

1. Build the design matrix row $\mathbf{x}_i$ (intercept + dummies for factor levels)
2. Compute the linear predictor: $\hat{\eta}_i = \mathbf{x}_i' \hat{\boldsymbol{\beta}}$
3. Apply the **inverse link**: $\hat{\mu}_i = g^{-1}(\hat{\eta}_i) = e^{\hat{\eta}_i}$

Here, SAS already gives us $\hat{\eta}_i$, so we just exponentiate:

$$\hat{\mu}_3 = e^{3.41} = 30.27, \quad \hat{\mu}_7 = e^{2.45} = 11.59$$

The log link makes effects multiplicative. Changing a factor level multiplies the expected claim by $e^{\hat{\beta}_j}$, regardless of the other factor levels.

Fitting GLMs: Maximum Likelihood

We find $\hat{\beta}$ by maximising the log-likelihood:

$$\ell(\beta) = \sum_{i=1}^n \left[\frac{y_i \theta_i - b(\theta_i)}{\psi} + c(y_i; \psi) \right]$$

Where is β ? It's hiding inside θ_i . Each θ_i depends on the mean μ_i , which depends on the linear predictor $\eta_i = \mathbf{x}_i' \beta$:

$$\beta \longrightarrow \eta_i = \mathbf{x}_i' \beta \longrightarrow \mu_i = g^{-1}(\eta_i) \longrightarrow \theta_i$$

The process: take $\partial \ell / \partial \beta_j$, set each to zero, and solve. In general, these equations have no closed-form solution.

In practice

`glm(y ~ x1 + x2, family = Gamma(link="log"))` handles everything.

Deviance: The GLM Version of RSS

First, some notation. $\ell(\hat{\boldsymbol{\mu}})$ is the log-likelihood of your fitted model, where $\hat{\boldsymbol{\mu}}$ is the vector of fitted values. Two benchmark models:

Saturated model (the “perfect model”): uses **one parameter per observation**, so $\hat{\mu}_i = y_i$ exactly. It memorises the data perfectly. Its log-likelihood $\ell(\mathbf{y})$ is the highest achievable score.

Null model (the simplest model): uses **only an intercept**, so $\hat{\mu}_i = \bar{y}$ for everyone. Same prediction for every observation.

Deviance measures how far your model is from the saturated model:

$$D = 2\psi[\ell(\mathbf{y}) - \ell(\hat{\boldsymbol{\mu}})]$$

$D \geq 0$, and smaller is better. For Normal with identity link, $D = \text{RSS}$.

Using Deviance: Testing Nested Models

If a simpler model is nested inside a bigger model, the drop in deviance tells you whether the extra variables are worth it.

Rule of thumb:

$$D_{\text{small}} - D_{\text{big}} > 2 \times (\text{extra parameters}) \implies \text{keep them}$$

For a formal test (when ψ is known, e.g. Poisson): compare $D_{\text{small}} - D_{\text{big}}$ to $\chi^2_{\Delta \text{d.f.}, 1-\alpha}$.

For **non-nested** models, use AIC instead (lower is better).

*R reports both null and residual deviance in `summary(glm(...))`.
The drop between them shows how much your predictors helped.*

Q4: Poisson GLM for French Auto Insurance

Pull this question up on your side. Here are the key hints:

- (a)** Rewrite $f(y) = e^{-\mu} \mu^y / y!$ as one $\exp(\dots)$
 $\rightarrow \theta = \ln \mu, \quad b(\theta) = e^\theta, \quad \psi = 1$
- (b)** Poisson log-likelihood: $\ell = \sum [y_i \ln \mu_i - \mu_i - \ln(y_i!)]$
Plug in $\mu_i = y_i$ (saturated) and $\mu_i = \hat{\mu}_i$ (fitted), subtract
The $\ln(y_i!)$ cancels
- (c)** Apply rule of thumb to the SAS deviance table
SEX and DRIVER_AGE: significant
VEH_AGE and LOYALTY: not significant
- (d)** Overdispersion: we'll cover this in Applied Q2

Pseudo- R^2

$$R^2_{\text{pseudo}} = 1 - \frac{D_M}{D_0}$$

Measures how much of the **null deviance** is explained by your model. Same idea as $R^2 = 1 - \text{RSS}/\text{TSS}$ in OLS.

- ▶ $R^2_{\text{pseudo}} = 0$: your model is no better than predicting $\bar{y}$ for everyone
- ▶ $R^2_{\text{pseudo}} = 1$: your model fits as well as the saturated model

Don't over-interpret

Pseudo- R^2 values for count and binary data are typically much lower than what you'd see in OLS. A value of 0.2–0.4 can be a perfectly good model. Don't compare it directly with an OLS R^2 .

Residuals for GLMs

Pearson residuals:

$$r_i^P = \frac{y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}$$

Raw residual divided by the standard deviation implied by the variance function.

Deviance residuals:

$$r_i^D = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$$

where d_i is observation i 's contribution to the total deviance.

Deviance residuals tend to be more normally distributed, so they're the default for diagnostic plots. Plot them against fitted values to check for patterns, just like in OLS.

Overdispersion (Preview)

Poisson assumes $\text{Var}(Y) = \mu$. If the empirical variance in the data is **much larger** than the mean, the Poisson model fails.

We'll see a concrete example in Applied Q2, where $\hat{\psi} \approx 55$ (the variance is $55\times$ what Poisson predicts). The fix: use **Negative Binomial** instead.

Applied Q2: Overdispersion in Practice

Third-party insurance claims data (claims vs accidents):

Model	AIC	Resid. Dev.	Issue
lm()	–	(RSS)	Non-Normal residuals
Poisson GLM	9796	8567	$\hat{\psi} = 55.4$
Neg. Binomial	2024	190	Much better fit

The Poisson model has $\hat{\psi} \approx 55$: the observed variance is $\sim 55 \times$ the mean. This is extreme.

Switching to Negative Binomial: AIC drops from 9796 to 2024, residual deviance from 8567 to 190, and diagnostic plots improve dramatically.

Quasi-Poisson gives the same $\hat{\beta}$ as Poisson, but inflates SEs by $\sqrt{55.4} \approx 7.4$. It doesn't produce an AIC though, so you can't use it for model comparison.

Thanks for coming! See you next week.

